Datathon 2026 Round 1: Growth Quality and Revenue Forecasting for Vietnamese Fashion E-commerce

PlotTwist

Track: EDA and Forecasting
https://github.com/tuanbachngo/Vin-Datathon

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Abstract

This report examines the quality of growth in a Vietnamese fashion e-commerce business. The key finding is that the company is still expanding in volume, but the economic quality of that growth is weakening. Promotion-heavy sales account for a meaningful share of quantity sold, yet promoted lines contribute a smaller share of net revenue and generate weak gross profit. At the same time, sessions, unique visitors, and signups continue to increase, while orders, revenue, and gross profit per 1,000 sessions decline over time. These patterns suggest that the business is not facing a simple traffic or volume problem, but a broader growth-quality problem. We recommend shifting management focus from volume-based KPIs toward profit-quality controls, including incremental gross profit from promotions, gross profit guardrails, and profitable conversion per session.

1 Exploratory Data Analysis

1.1 Promotion Prevalence and Gross Margin Pressure

1.1.1 Promotion-driven volume has weak profit quality

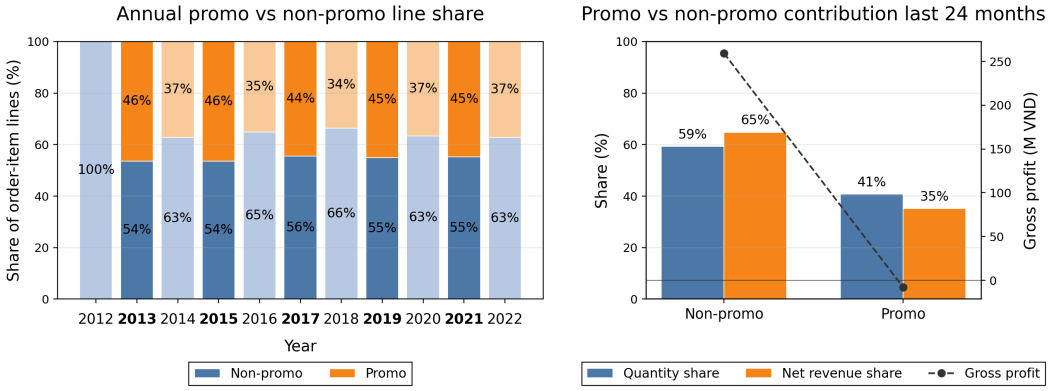


Figure 1: Promotion prevalence and profit quality of promoted sales volume.

Interpretation of Figure 1. Promotion remains highly prevalent in order-item lines, especially in odd years, where promo line share reaches roughly 44–46%. This suggests a recurring biennial promotion pattern, meaning the promotion–margin relationship may partly reflect a seasonal or strategic campaign cycle rather than only the direct effect of discounting. In the last 24 months, promoted items account for around 41% of quantity sold but only 35% of net revenue, while promo gross profit is close to zero or negative. This indicates that discounting helps generate volume, but the volume created by promotions has weak profit quality.

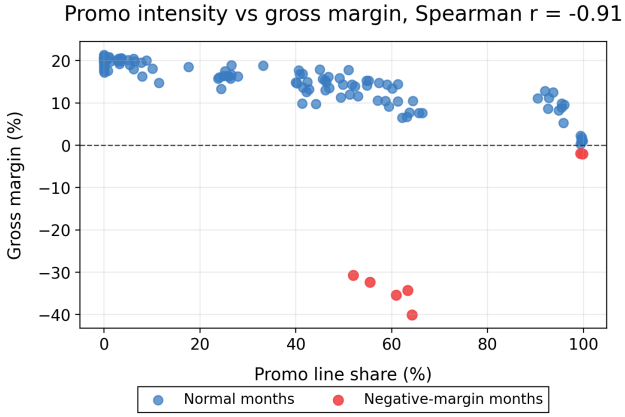


Figure 2: Monthly relationship between promo line share and gross margin.

Interpretation of Figure 2. At the monthly level, promo line share has a very strong negative relationship with gross margin, with Spearman $r_s = -0.91$. Negative-margin months are concentrated in high-promotion periods, suggesting that promotion-heavy months are tightly associated with margin pressure.

1.1.2 Risk if the business becomes dependent on promotions

If the current pattern continues, promotions may train customers to wait for discounts rather than buying at full price. In that case, promotions stop being a temporary demand stimulus and become a requirement for maintaining sales volume. Similar cases in fashion retail show that excessive discounting can create sale-seeking behavior and compress profit margins.

The key risk is a discount-dependency trap: reducing promotions may hurt short-term sales, while continuing promotions keeps gross margin under pressure. In our data, promoted items account for a large share of quantity sold but contribute a smaller share of net revenue, with promo gross profit close to zero or negative. This suggests that promotion-heavy growth may protect visible sales volume while weakening the economic quality of that volume. Therefore, future promotion planning should monitor not only sales uplift, but also full-price demand, promo revenue efficiency, and incremental gross profit.

1.1.3 Managerial recommendation: govern promotion by gross profit

The business should shift promotion KPIs from sales volume or revenue uplift to incremental gross profit. The objective is not to eliminate all promotions, but to prevent promotions from creating volume that does not contribute to profit.

- **Set gross profit guardrails with holdout tests.** Before launch or expansion, a promotion should pass two conditions: projected promo gross profit is positive, and expected incremental gross profit is above zero relative to a matched non-promo baseline. Holdout groups should then be used to validate whether the campaign creates additional gross profit or simply shifts full-price demand into discounted sales.
- **Optimize discount depth by margin thresholds.** Instead of applying one discount level broadly, the business should test several discount depths and choose the level that maximizes incremental gross profit rather than sales volume.
- **Use promo revenue efficiency as an expansion rule.** Before expanding a promotion-heavy campaign, the business should compare the promo revenue share with the promo quantity share:

$$\text{Promo revenue efficiency} = \frac{\text{Promo revenue share}}{\text{Promo quantity share}}.$$

  A campaign should only be expanded when this ratio is close to or above 1, meaning promoted volume monetizes at least as well as the overall average. If the ratio is materially below 1, the campaign should be reviewed before scaling.

1.2 Traffic Growth but Weakening Commercial Efficiency

1.2.1 Traffic grows, but monetization efficiency declines

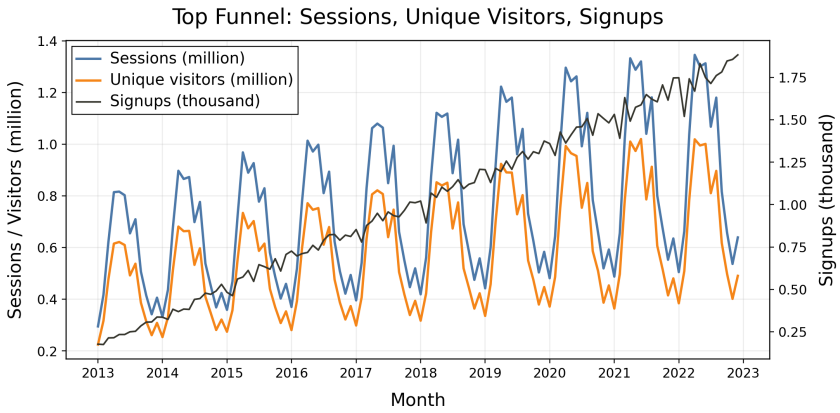
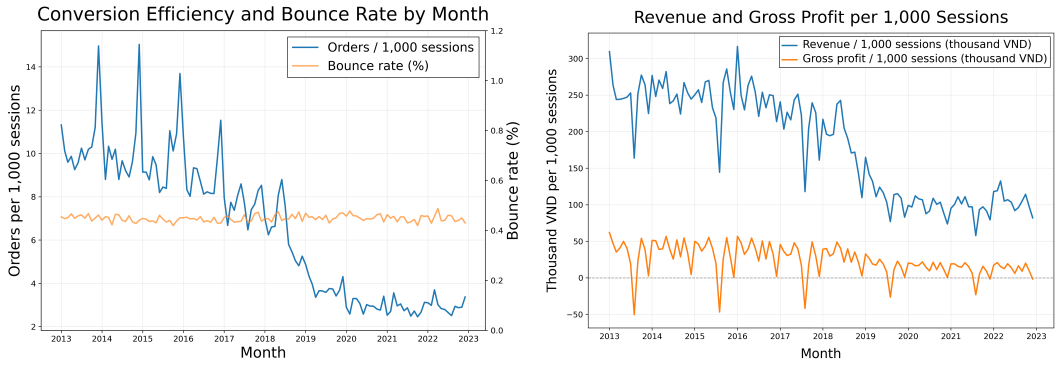


Figure 3: Top-of-funnel development over time: sessions, unique visitors, and signups.

Interpretation of Figure 3. Sessions, unique visitors, and signups all increase over time, showing that the company is still able to expand its top-of-funnel. However, traffic growth must be read together with conversion and commercial efficiency, because more traffic does not automatically translate into more orders or higher profit.



(a) Orders per 1,000 sessions and bounce rate.

(b) Revenue and gross profit per 1,000 sessions.

Figure 4: Traffic monetization efficiency declines despite stable bounce rate.

Interpretation of Figure 4. Panel (a) shows a sharp decline in orders per 1,000 sessions while bounce rate remains broadly stable, suggesting that the conversion bottleneck is likely deeper in the post-click funnel rather than at the landing stage. Panel (b) shows that revenue and gross profit per 1,000 sessions also decline, indicating that each unit of traffic generates less commercial value over time. Together, the two panels show weaker conversion efficiency and deteriorating traffic monetization quality.

1.2.2 Risk if seasonal traffic peaks become less monetizable

If the current trend continues, future traffic peaks may look strong in volume but weak in commercial value. This is a realistic e-commerce risk: industry benchmarks show that conversion rates can decline even when traffic acquisition continues, while cart abandonment and checkout friction remain major barriers to completed purchases. For this business, the key risk is that seasonal sessions may increase without proportional growth in orders, revenue, or gross profit. Therefore, peak-season planning should not forecast sessions alone, but should also monitor orders, revenue, and gross profit per 1,000 sessions to distinguish healthy demand from low-quality traffic growth.

1.2.3 Managerial recommendation: optimize traffic for profitable conversion

The management focus should shift from maximizing traffic volume to improving commercial efficiency per session.

- **Shift acquisition KPIs from volume to profitable conversion.** Instead of optimizing only sessions, visitors, or signups, the business should evaluate each traffic source using orders, revenue, and gross profit per 1,000 sessions. This helps distinguish high-volume sources from high-value sources.
- **Diagnose post-click funnel quality by source and landing page.** Since bounce rate remains stable while conversion efficiency declines, the business should monitor deeper funnel metrics such as product-page engagement, add-to-cart rate, cart abandonment, payment completion, and checkout friction. Future tracking should evaluate these metrics by source–landing-page pair to identify which acquisition paths create profitable conversion rather than only visits.

1.3 Integrated Business Implication

The two EDA findings point to the same underlying issue: growth volume is becoming less profitable. Promotion-heavy sales volume shows weak profit quality, while growing traffic produces fewer orders, lower revenue, and lower gross profit per 1,000 sessions. Therefore, the company should manage growth quality through three connected metrics:

- **Incremental gross profit from promotion:** to ensure that discounts create additional profit rather than only additional volume.
- **Gross profit per 1,000 sessions:** to evaluate whether traffic growth creates economic value.

- **Profitable conversion by source:** to prioritize acquisition paths that generate orders and gross profit, not just visits.

2 Revenue Forecasting Model

This section summarizes the model used to forecast daily revenue from 2023-01-01 to 2024-07-01. The target is `Revenue`; the training data come from `sales.csv`, covering 2012-07-04 to 2022-12-31. The selected submission is a pre-Covid anchor XGBoost model with residual learning, isotonic calibration, and a final `x115` multiplier under the tag `low_w2020_2022`.

2.1 Modeling approach

The model does not forecast raw revenue directly. It first builds a seasonal revenue baseline and then predicts the remaining log-residual. This design is useful for long-horizon daily forecasting because revenue contains strong calendar seasonality, holiday effects, and possible regime shifts after Covid. The default residual anchor is mainly the same-calendar-date 1–2 year baseline, with lookup fallback when needed. Growth-adjusted and `month` $\times$ `day-of-week` lookup baselines are included as residual-baseline features, gaps, and ratios, allowing XGBoost to learn when to deviate from the active anchor. A pre-Covid anchor from 2017-2019 is added to recover a more stable seasonal shape and reduce dependence on abnormal 2020-2022 patterns.

2.2 Features and training

Each forecast date uses calendar and holiday variables, seasonal lookup values, residual-baseline variables, forecast-safe auxiliary/commercial seasonality variables, and pre-Covid anchor-gap variables. The pre-Covid component includes monthly means, `month` $\times$ `day-of-week` means, Tet-bucket means, and Q4 means calculated from 2017-2019. These values are converted into gaps and ratios against the active baseline so that XGBoost can decide when to follow the pre-Covid seasonal pattern and when to follow the recent revenue level.

The training target is the log residual, $\log(\texttt{Revenue}) - \log(\texttt{baseline_anchor})$. XGBoost is trained with a conservative tree setup: learning rate 0.025, 650 trees, `max_depth` $= 3$, `min_child_weight` $= 10$, `subsample` $= 0.8$, `colsample_bytree` $= 0.65$, `gamma` $= 0.1$, `reg_alpha` $= 1$, and `reg_lambda` $= 10$. The weighting scheme gives weight 2.0 to 2017-2019 and weight 0.5 to both 2012-2016 and 2020-2022, reflecting the assumption that the pre-Covid seasonal structure is most reliable for the test horizon.

2.3 Validation and final choice

Walk-forward validation uses three folds: `fold1_2020H2-2021`, `fold2_2021H2-2022`, and `fold3_test_proxy`. The final score applies weights of $0.2/0.3/0.5$, giving the largest weight to the fold closest to the forecast horizon.

Table 1: Cross-validation results after isotonic calibration.

Split	RMSE	R^2	MAPE
`fold1_2020H2-2021`	801,674	0.734	0.227
`fold2_2021H2-2022`	808,328	0.729	0.227
`fold3_test_proxy`	808,933	0.766	0.206
Weighted overall	**807,299**	**0.749**	**0.217**

Isotonic calibration improves weighted RMSE from $932{,}230$ to $807{,}299$ and raises weighted R^2 from 0.664 to 0.749. It also reduces daily aggregated RMSE from $951{,}165$ to $771{,}489$ and lowers peak-month MAPE from 15.70% to 10.03%. The final `x115` multiplier is leaderboard-oriented, but it is consistent with the validation pattern that the calibrated pre-Covid residual model performs best among the tested variants.

From a business perspective, the model should be read as a seasonality-aware revenue planning tool. The pre-Covid anchor preserves a stable historical reference, while recent level corrections keep the forecast connected to the latest observed revenue scale.

A Appendix: KPI Definitions and Additional Validation Needs

A.1 Definitions of key KPIs

Table 2: Key KPIs used in the analysis.

KPI	Definition	Role in the analysis
Promotion-related KPIs		
Promo line share	Share of order-item lines with at least one promotion applied.	Measures the prevalence of promotions.
Promo quantity share	Share of sold quantity coming from promoted order-item lines.	Measures how much sales volume is promotion-driven.
Promo revenue share	Share of net revenue generated by promoted order-item lines.	Compares revenue contribution with volume contribution.
Promo gross profit	Gross profit generated by promoted order-item lines.	Measures the profit quality of promoted volume.
Incremental gross profit	Difference in gross profit between a promoted group and a comparable non-promoted or holdout group.	Measures whether a promotion creates additional profit rather than merely shifting full-price demand.
Traffic and conversion-efficiency KPIs		
Orders per 1,000 sessions	Number of non-cancelled orders generated per 1,000 sessions.	Measures the efficiency of converting traffic into orders.
Revenue per 1,000 sessions	Net revenue generated per 1,000 sessions.	Measures traffic monetization efficiency.
Gross profit per 1,000 sessions	Gross profit generated per 1,000 sessions.	Measures the economic quality of traffic.
Bounce rate	Share of sessions in which users leave after minimal interaction.	Helps identify whether users drop off immediately after landing.
Pages per session	Average number of pages viewed in a session.	Helps determine whether users continue browsing after landing.
Average session duration	Average time spent per session.	Helps evaluate engagement quality beyond bounce rate.

A.2 Additional hypotheses and validation needs

Table 3: Initial hypotheses that require further validation.

Initial hypothesis	Current assessment	Next validation step
Promotion is required to sell products.	Not yet proven.	Test full-price sell-through, promo holdout groups, and the extent to which promotions shift non-promo demand into discounted sales.
The company has lost pricing power.	Cannot be concluded yet.	Estimate price elasticity, compare with competitor prices, or run discount-depth experiments to evaluate ability to sell at full price.
Promotion is mainly used to clear inventory.	Not yet proven.	Test the relationship between inventory pressure, stockout/overstock signals, and the probability of running promotion in the following period.
Promotion causes gross margin decline.	Strong association, not causal proof.	Control for COGS changes, category mix, product mix, seasonality, and campaign timing; ideally run holdout tests.
Odd-year promotion intensity is a recurring business pattern.	Supported descriptively, but mechanism unclear.	Compare odd-year and even-year promo intensity, margin, product mix, and campaign timing. Additional competitor or market-cycle data is needed to determine whether the pattern is strategic or externally driven.
Organic traffic has post-click mismatch.	A hypothesis to monitor.	Add search query, landing page, session ID, and source–page conversion tracking to test user intent and post-click experience.
Paid/social traffic targeting is inefficient.	Insufficient data to conclude.	Add campaign spend, CAC, CPC, and gross profit after acquisition cost to evaluate targeting quality and ROI.
Low rating reduces conversion.	Not clearly supported at monthly aggregate level.	Analyze at product or landing-page level, especially for high-traffic products with high low-rating share.
Conversion decline is driven by immediate bounce.	Not supported by current bounce-rate pattern.	Analyze pages per session, average session duration, product-page views, add-to-cart rate, cart abandonment, and checkout completion rate.

A.3 Data limitations and recommended future tests

The conclusions in the main report are based primarily on monthly aggregates, traffic-source level data, and order-item level data. Therefore, the results should be interpreted as evidence of patterns and associations, not as complete causal proof.

To validate the mechanisms more rigorously, the business should implement or collect the following:

- **Promo holdout test:** compare promoted SKUs or customer groups with similar non-promoted holdout groups to estimate incremental gross profit.
- **Discount-depth test:** test different discount levels to identify the level that maximizes gross profit rather than revenue or volume.
- **COGS and product-mix decomposition:** separate margin pressure into discounting effects, COGS effects, and product/category mix effects.
- **Seasonality control:** explicitly compare odd-year and even-year promotion intensity and gross margin to avoid mistaking recurring seasonal patterns for pure causal effects.
- **Landing-page tracking:** add session ID, landing page, and page-level behavior to test post-click mismatch.
- **Post-click funnel metrics:** collect product-page view rate, add-to-cart rate, cart abandonment rate, payment failure rate, and checkout completion rate.
- **Customer cohort analysis:** measure signup-to-first-order, first-to-second-order rate, repeat purchase rate, and gross profit per customer by cohort.
- **Campaign spend data:** add spend, CPC, CAC, and campaign budget to evaluate acquisition efficiency after marketing cost.

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